

interacting with and studying plants difficult. Combining plants with electronic functionality would make it possible to combine electric signals with the plant's own chemical processes. Controlling and interfacing with chemical pathways in plants could pave the way to photosynthesis based fuel cells, sensors and growth regulators, and devices that modulate the internal functions of plants.

These are some steps taken to merge the fields of organic electronics and plant science. The aim is to develop applications for energy, environmental sustainability and new ways of interacting with plants.

Device to harvest energy from walking

Researchers have demonstrated a universal self-charging system driven by random body motion. By walking or running, the system uses the tribo-electric effect to generate electricity by physically manipulating two materials with opposing surface charges. The current is enough to power sensors, microcontrollers, memories, arithmetic logic units, displays and even wireless transmitters.

More complicated electronics like smartwatches, mobile phones, navigation systems, tablets, personal computers and sensor nodes in the Internet of Things can also be charged using the same system if high-frequency mechanical agitations are utilised to drastically improve tribo-electric nanogenerator's (TENG's) output power.

To embed TENGs into an electronic system for practical applications, the team devised a system that includes a TENG, a power-management circuit and a low-leakage energy-storage device. The power management circuit is designed to solve the impedance mismatch problem. It can achieve 90 per cent board efficiency and 60 per cent total efficiency, about two orders of magnitude improvement compared with direct charging.

Sensor that can detect oestrogen levels in body fluids

Oestrogen is one of the main hormones that regulates the female reproductive system. Researchers at Victoria University of Wellington, New Zealand, have developed a sensor that can detect low levels of E2, one of the primary oestrogen hormones, in liquids like saliva.

The sensor sends an electronic signal in the presence of oestrogen and, with further development, can test oestrogen levels in body fluids or test waterways for oestrogen contamination that might pose a risk to humans and the environment.

It has a simple design, gives real-time readings, could be integrated into an electronic monitoring system and uses very little power. It uses small snippets of DNA called aptamers to latch onto oestrogen molecules.

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